MES ASSIGNMENT CA-1

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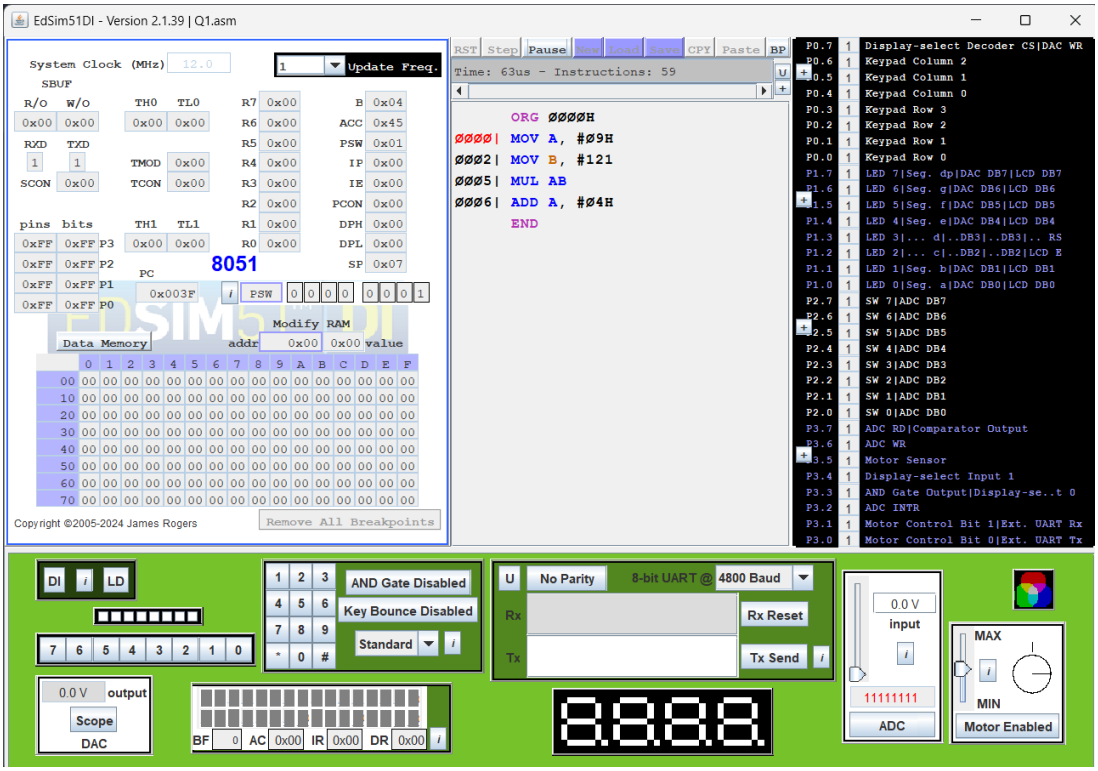
SEM: IV, B, 2024-28

Q1. Write an 8051 Assembly Language Program (ALP) to generate the last four digits of your PRN using any arithmetic instructions. The program should not directly load the complete PRN number as an immediate value. Instead, it must use appropriate arithmetic operations such as ADD, MUL, or INC to form the number logically. The final result must be stored in the Accumulator register (AX). For example, if a student's PRN is 24070521091, the last four digits are 1091, and the value 1091 should be available in AX at the end of program execution.



The program constructs the number 1093 step by step using arithmetic operations instead of loading it directly. Initially, two small values are added in the accumulator to produce the number 9. This value is then multiplied by 126 using the MUL AB instruction. Since the multiplication results in a 16-bit output, the lower byte is stored in the Accumulator while the higher byte is stored in register B. After that, an additional value of 5 is added to the accumulator to obtain the final result 1093. Hence, the required last four digits are generated logically and the complete result is present in the AX (A and B registers).

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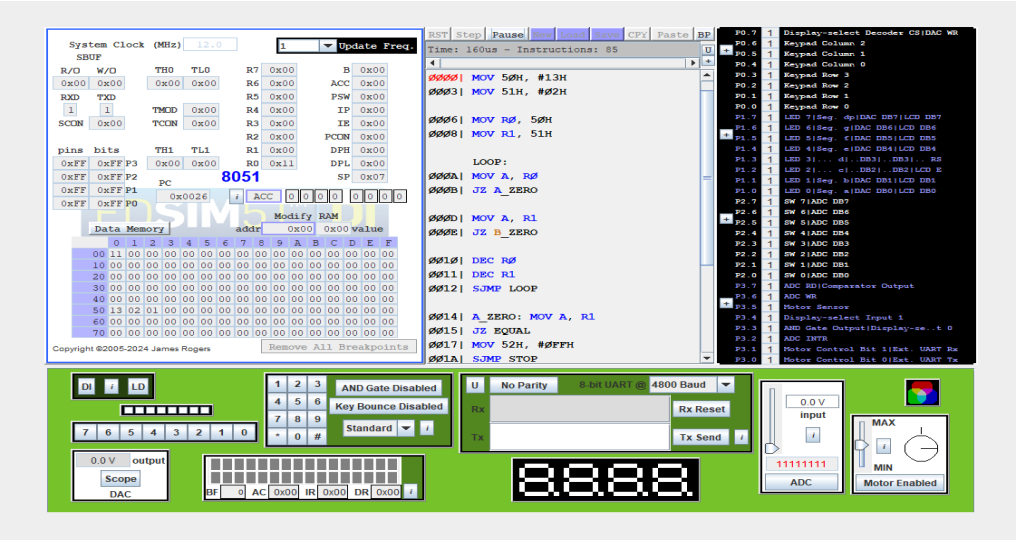
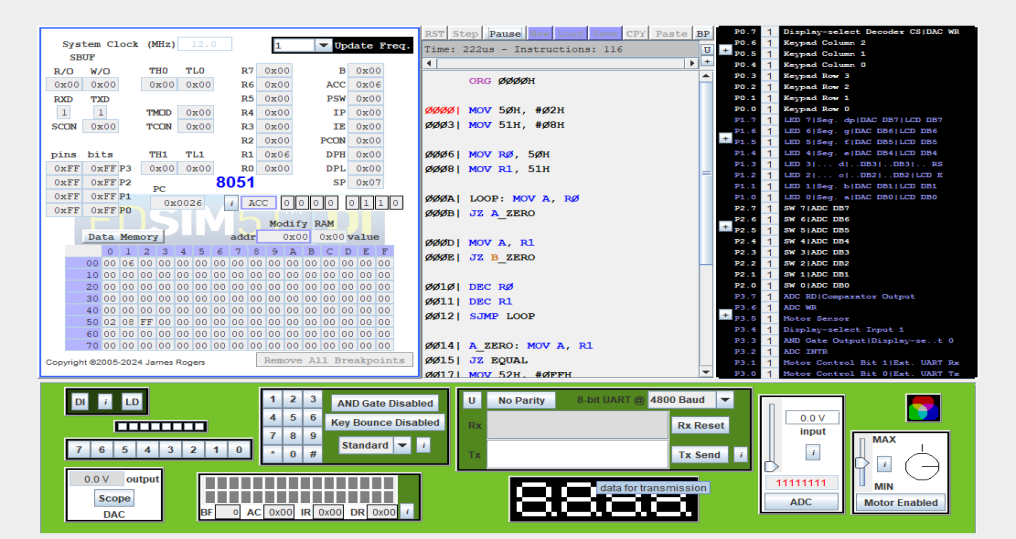
Q.2 Execute an 8051-assembly language program for a safety-certified system in which the instructions CJNE, DJNZ, and SUBB are not permitted. Two unsigned numbers are stored in internal RAM locations 50H and 51H. The program must compare these two numbers using only the allowed instruction set (MOV, INC, DEC, JZ, JNZ, CLR, SETB, ANL, ORL) and store the comparison result in a register or memory location such that 01H indicates the value at 50H is greater than the value at 51H, 00H indicates both values are equal, and FFH indicates the value at 50H is less than the value at 51H. The program should be simulated for all three possible cases (A > B, A = B, A < B), and the solution must clearly explain how flag behavior (especially the Zero flag) is utilized to achieve comparison under the given instruction constraints.

For A>B



For A<B



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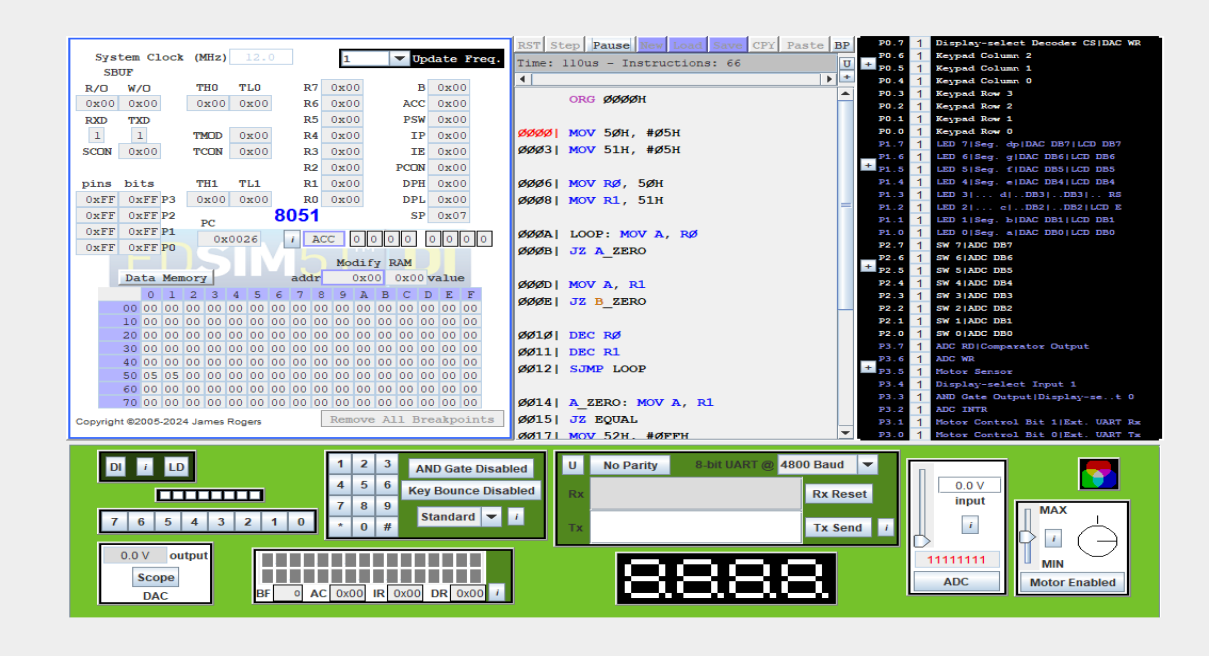
For A=B



In this program, two unsigned numbers stored at memory locations 50H and 51H are compared without using the instructions CJNE, DJNZ, and SUBB. The values are first moved into registers R0 and R1. Both registers are then decreased together in a loop. After each step, the Zero flag is checked using the JZ instruction. If R1 becomes zero first while R0 is still not zero, the program stores 01H in memory location 52H, which means A > B. If R0 becomes zero first while R1 is still not zero, the program stores FFH in location 52H, which means A < B. If both R0 and R1 become zero at the same time, the program stores 00H in location 52H, which means A = B. In this way, the comparison is done using only the allowed instructions and by using the Zero flag.

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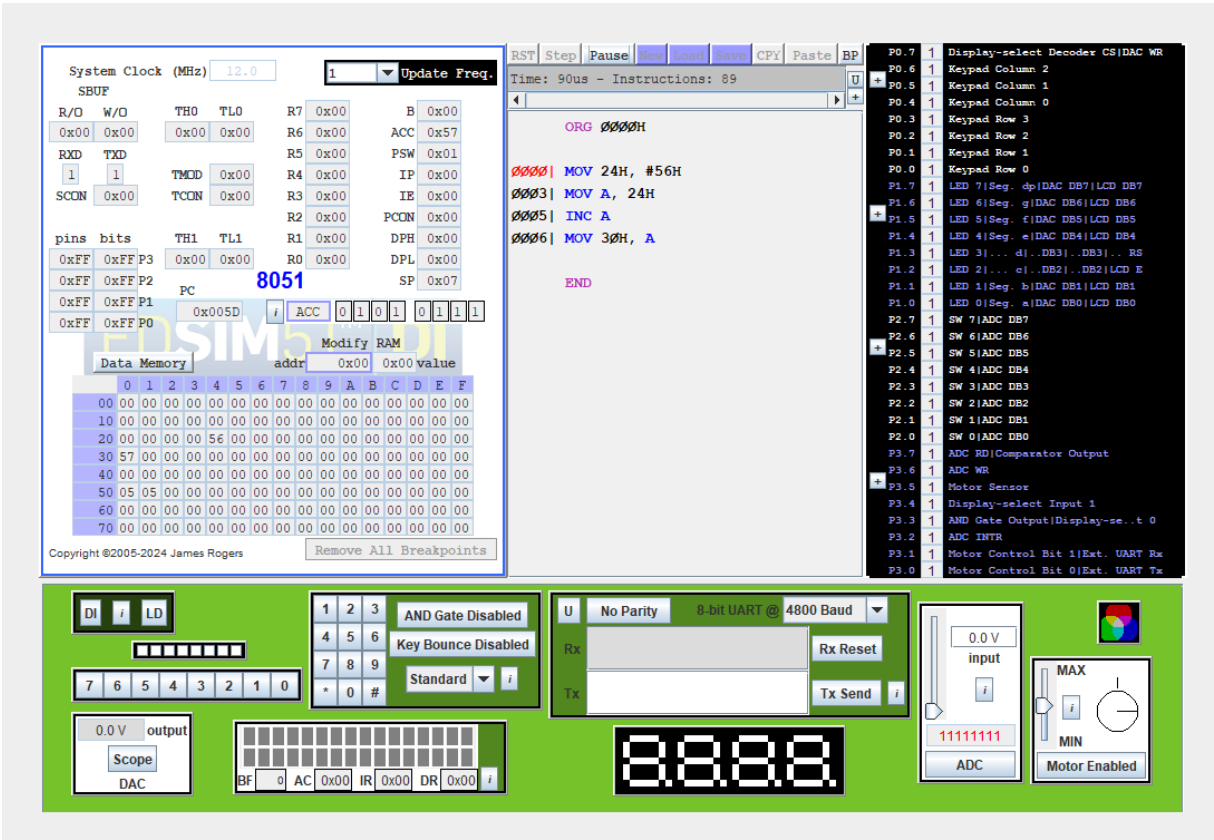
Q.3 A student claims that two assembly programs are equivalent because both access the same RAM address; however, this claim is incorrect due to the difference in addressing modes. In this case study, write two short assembly programs—one using direct addressing and the other using indirect addressing—such that both reference the same RAM location. Using an appropriate initial RAM configuration, demonstrate a situation where the outputs of the two programs differ even though the base address is the same. Support the observation with register and RAM snapshots from simulation, and explain that the difference arises because direct addressing accesses the data stored at the given address, whereas indirect addressing treats the contents of that address as a pointer to another memory location, leading to different data being fetched and hence different outputs.



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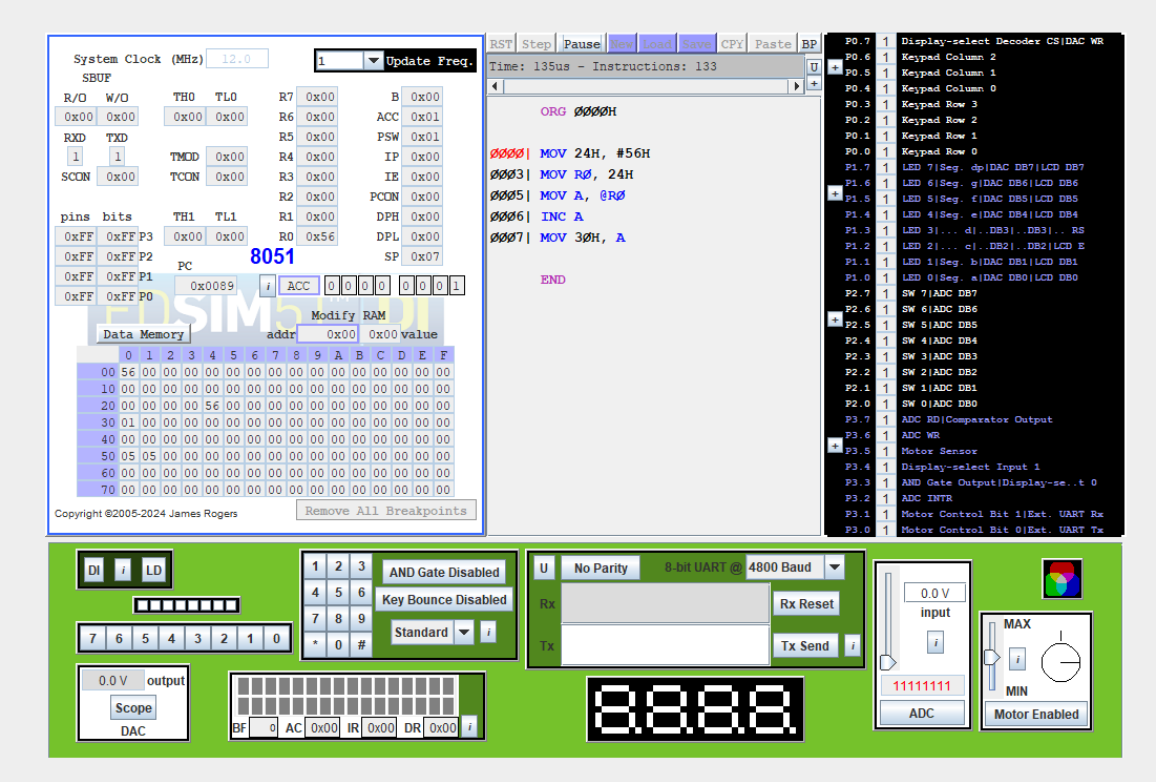

Two programs access the same RAM address (24H) using different addressing modes. In direct addressing, MOV A, 24H reads the data stored at 24H (56H) and after incrementing, stores 57H at 30H. In indirect addressing, MOV R0, 24H treats the value at 24H as a pointer and MOV A, @R0 reads data from the memory location pointed to by 56H. Hence, even though both programs use address 24H, their outputs differ because direct addressing fetches data directly from that address while indirect addressing fetches data from another location.

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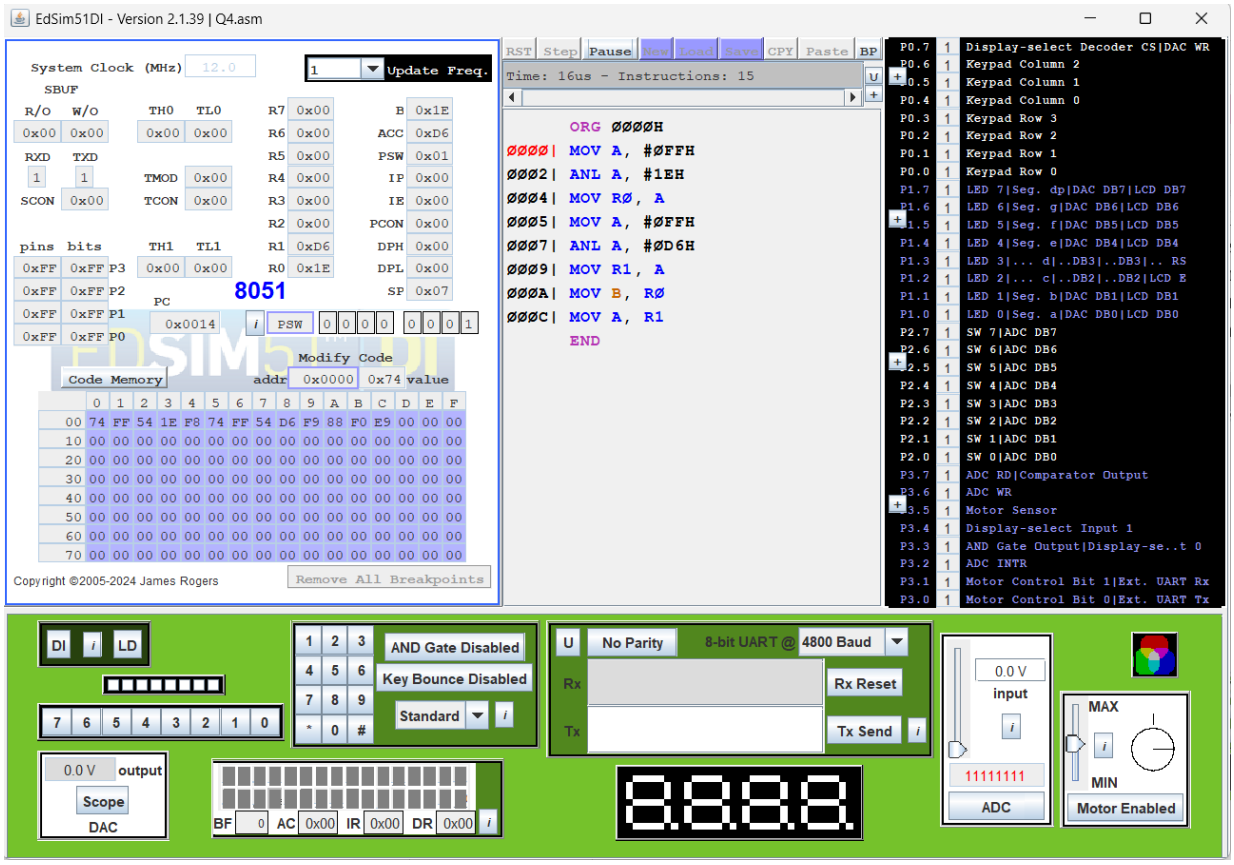
Q.4 Write an 8051 Assembly Language Program in which you must use logical instructions to construct a numeric result. Using multiple logical instructions such as ANL, ORL, and CLR, generate the last four digits of your own mobile number through a suitable sequence of operations (you may split the digits and combine them logically as required). Do not directly load the complete 4-digit number as an immediate value. The program should use more than one logical instruction, and at the end of execution the Accumulator (A) must contain the last four digits of your mobile number. Simulate the program and verify that the final value in the Accumulator matches your mobile number's last four digits.



In this program, the last four digits of the mobile number (7894) are generated using logical instructions instead of directly loading the value. The number is divided into a high byte and a low byte. First, the accumulator is loaded with FFH and masked using the ANL instruction to form the high byte (1EH). This value is stored in register R0. Then, the accumulator is again loaded with FFH and masked using ANL to form the low byte (D6H), which is stored in register R1. Finally, the high byte is moved to register B and the low byte is moved to the accumulator. Thus, the combined value in the AX (B:A) register pair represents 7894, verifying that the required last four digits are generated using logical instructions.

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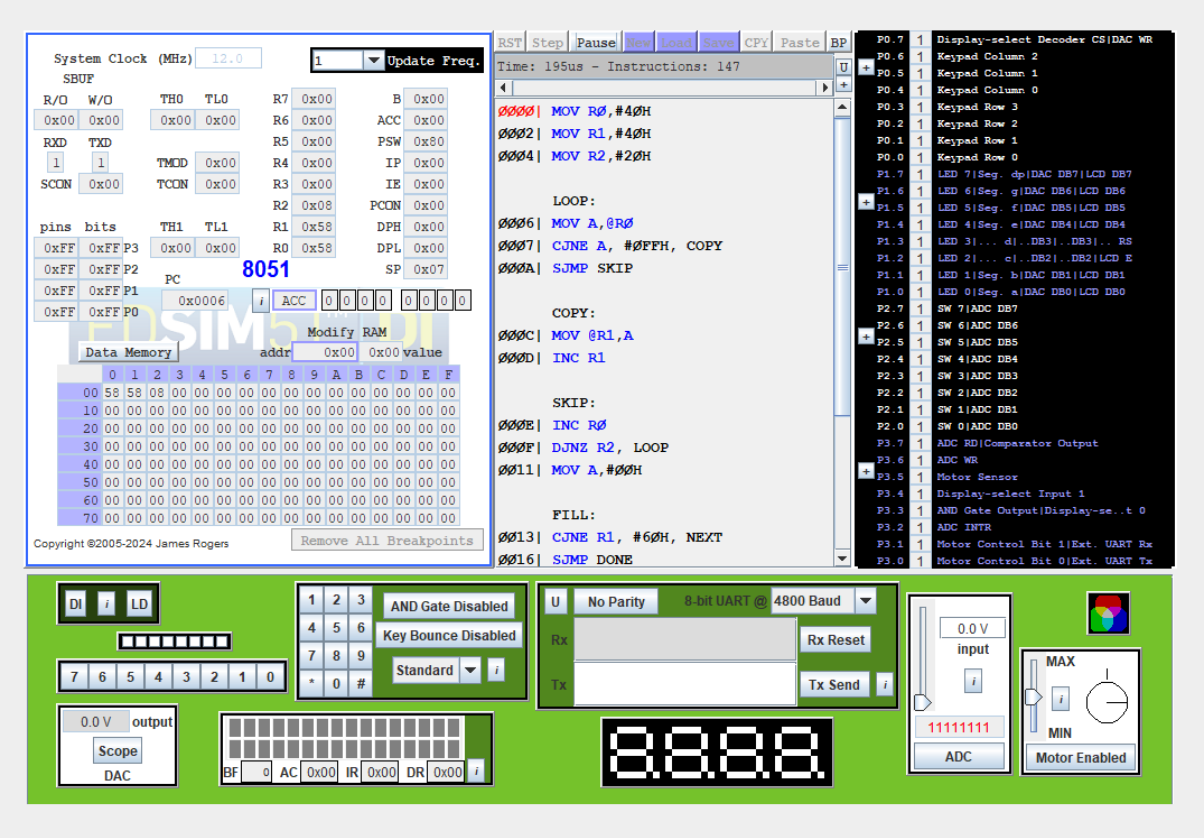
Q.5 An embedded logger stores event codes in internal RAM from 40H to 5FH, but due to strict memory limitations the data must be compacted in-place without using any additional RAM or the stack. Write an assembly language program that scans the memory range 40H–5FH using only indirect addressing, removes all occurrences of the value FFH, shifts the remaining valid data bytes to the left to eliminate gaps, and fills the unused memory locations at the end of the range with 00H. **Execute** the program to show the RAM contents before and after execution, and clearly explain the pointer movement logic used to identify valid data, shift it correctly, and overwrite invalid entries under the given constraints.



This program scans the internal RAM from address 40H to 5FH using indirect addressing. The read pointer (R0) checks each memory location and compares the data with FFH. If the value is not FFH, it is copied to the location pointed by the write pointer (R1), thereby shifting valid data to the left and removing gaps. After all bytes are processed, the remaining memory locations are filled with 00H. Thus, all FFH values are removed, valid data is compacted, and unused locations are cleared.

Github repo link - https://github.com/Harsh2227kumar/MES-CA-I/